

Aditya Pratap Singh Tomar

B.Tech. Computer Science Engineering · Jabalpur Engineering College

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ACHIEVEMENTS

- **GATE 2026 – Top 5%** in Data Science & AI (DS&AI): AIR 3930 out of 80,000+ candidates nationwide.
- **Kaggle Top 1.7%** – Rank 196 of 11,450 participants, Playground Series S5E11 (tabular ML competition).
- **SIH 2025 Finalist** – Smart India Hackathon, Problem Statement PS-25189 (national government-backed hackathon).
- **HackCrux Runner-up** – National-level hackathon, LNMIIT Jaipur; competitive ML/systems track.

EDUCATION

- **Jabalpur Engineering College** Sep 2023 – Apr 2027
B.Tech., Computer Science Engineering – CGPA: **8.39 / 10.0** Jabalpur, MP
- **CBSE Senior Secondary (Class XII)** 2023
83.4% - Physics, Chemistry, Mathematics, Computer Science, English
- **CBSE Senior Secondary (Class X)** 2021
93% - Mathematics, Science, English, Social Science, Hindi

EXPERIENCE

- **Messgerat Labs Pvt. Ltd.** Jun 2026 – Present
SDE Intern Indore, MP
 - Architecting and developing a scalable EdTech/LMS platform using React, FastAPI, PostgreSQL, Redis, and JWT authentication, enabling secure role-based learning workflows and modular service design.
 - Designing and implementing production-grade backend systems including REST APIs, RBAC, course delivery pipelines, caching layers, database optimization, and media streaming integrations for high-availability applications.
 - Deploying and optimizing cloud-native infrastructure with Docker, CI/CD automation, monitoring, and performance-focused system architecture, supporting reliable and scalable production releases.
- **Cronlabs Solutions Pvt. Ltd. (Onetapp Group)** Feb 2026 – May 2026
AI Engineer Remote
 - Engineered LangChain/LangGraph agentic pipelines for planned resume customization on a Django + FastAPI SaaS backend, cutting time-to-tailored-resume **94% (25 min → 90 sec)** for **500+ active users**; owned end-to-end CI/CD deployment, production monitoring, and weekly Agile release cycle.
 - Serviced semantic resume-job matching system using sentence-transformer embeddings and ATS keyword-gap scoring over **3,000+ daily job postings**; integrated REST API endpoints and N8N workflow orchestration, improving candidate-role relevance **~60% vs. keyword-only baseline**.
- **CatalyzeIQ Pvt. Ltd.** Dec 2025 – Feb 2026
Lead Engineer Hybrid
 - Architected distributed backend system from zero-to-production for a B2B SaaS platform: Redis caching reduced API latency by **35%**, async task execution handled **50+ concurrent requests**, microservice architecture enabled independent deployment across 4 services; set up Jenkins CI/CD on EKS System.
 - Defined Prometheus + Grafana observability stack with alerting thresholds, reducing mean time-to-detection (MTTD) by **~40%**; planned gRPC/REST API conventions and integration testing strategy enabling zero-downtime weekly release cadence throughout early-stage iteration.
- **Suryavanshi Ventures Pvt. Ltd.** Aug 2025 – Sep 2025
AI Research Intern Hybrid
 - Developed production RAG pipelines (LangGraph + LangChain + FAISS vector store) with domain-specific document chunking and hybrid retrieval; improved healthcare agent factual accuracy **~30% over vanilla LLM baseline** on 500+ clinical queries, grounding responses in clinical context.
 - Reduced FastAPI backend mean latency **20% (850ms → 680ms; p95 improved 35%)** across **10,000+ daily API requests** via async refactoring and SQLAlchemy query indexing; structured MLOps data versioning (DVC) and feature store schema for production healthcare ML workflows.

SKILLS

- **Languages**: Python, Java, C/C++, SQL
- **Backend & Systems**: FastAPI, Flask, Django, REST APIs, gRPC, Microservices, Distributed Systems, Async Python, Redis, Linux, System Design
- **Data Science & ML**: PyTorch, TensorFlow, Scikit-Learn, XGBoost, HuggingFace Transformers, Transformer Architecture, NumPy, Pandas, SciPy, A/B Testing, Model Evaluation, Benchmarking
- **MLOps & DevOps**: MLflow, DVC, Feature Stores, ONNX, Docker, Kubernetes, Airflow, AWS, GitHub Actions, Prometheus, Grafana, CI/CD, Data Pipelines, Drift Detection
- **GenAI & Agentic AI**: LangChain, LangGraph, RAG, Vector Databases (FAISS, Chroma), OpenAI API, Gemini API, Prompt Engineering, Production ML, Agentic Workflows
- **Practices**: Agile/Scrum, API Design, Observability, Cross-functional Collaboration, Integration Testing, Technical Mentorship

MAJOR PROJECTS

- **ForesightX – Production MLOps Platform for Market Forecasting** Jan 2026
Python, PyTorch, MLflow, DVC, Docker, Kubernetes, gRPC, Redis, ONNX, GitHub Actions [GitHub, Live]
 - Engineered Temporal Fusion Transformer achieving **4.2% MAPE on 30-day equity forecasts** over 1M+ OHLCV records – **41% relative improvement vs. ARIMA baseline (7.1% MAPE)**; designed production feature store with 14+ technical indicators and DVC-tracked data versioning with MLflow experiment tracking.
 - Built microservice architecture with REST/gRPC serving, Redis response caching, and ONNX-quantized inference; reduced model serving latency **31% (420ms → 290ms at p95)** under 200 concurrent requests with Kubernetes-orchestrated training and serving workloads.
 - Deployed on AWS with GitHub Actions CI/CD, Prometheus + Grafana observability, automatic data drift detection and model retraining workflows, and integration test suite validating data pipeline, model serving, and API contracts end-to-end.
- **PulseCare AI – Intelligent Cardiac Diagnostics Platform** Jul 2025
Python, PyTorch, React, Flask, WebRTC, SciPy, NumPy, Scikit-learn, HuggingFace Transformers [GitHub, Live]
 - Trained transformer-based ECG classifier on MIT-BIH Arrhythmia Dataset (5 classes, 109k+ samples), achieving **91.3% macro-F1 vs. 84.1% ResNet-1D baseline** with sub-150ms CPU inference; benchmarked with per-class precision/recall and confusion matrix analysis.
 - Full-stack telehealth platform with WebRTC video consultations, secure patient record management, and real-time AI-assisted diagnostics (anomaly detection + rhythm classification) integrated into a production Flask + React architecture.
- **CareerPilot AI – Resume & Job Matching Platform** Feb 2026
Python, React, Flask, OpenAI API, Sentence-Transformers, Tailwind CSS, NLP, System Design [GitHub, Live]
 - Implemented semantic job-matching data pipeline via sentence-transformer cosine similarity over **3,000+ scraped postings** with LLM-driven resume rewriting, ATS keyword-gap analysis, and recruiter-focused scoring; complete pipeline runs end-to-end in under 2 minutes per candidate.
 - Delivered automated full-stack resume processing platform (React + Flask + OpenAI API) handling concurrent document analysis at sub-5s response latency; real-time feedback loop reduced average job-application preparation time from 45 min to under 5 min per candidate.
- *More projects at [apst.me/projects](#) – VisionInspect AI, IntelliHire AI, and others.*

CERTIFICATIONS

- **GPU Architectures and Programming**, [NPTEL](#) May 2026
- **Natural Language Processing**, [NPTEL](#) May 2025
- **CS50x**, [Harvard University](#) Nov 2024
- **Ethical Hacking**, [NPTEL](#) Dec 2024

POSITIONS OF RESPONSIBILITY

- **Tech-Team Lead**, MATRIX, JEC Jun 2024 – Present
 - Mentored 50+ member team for Winter-Arc initiative; drove 400+ participants through an AI-powered learning platform via cross-functional coordination, sponsorship acquisition, and full platform delivery.
- **Technical Member**, JLUG, JEC Oct 2024 – Present
 - Delivered DSA technical seminar to 100+ student developers; mentored juniors and supported coding community via peer workshops and event organization.